

- 1) $\cos(-\theta) = ?$
- a) $\cos\theta$
 - b) $-\cos\theta$
 - c) $\sin\theta$
 - d) $-\sin\theta$
- 2) The smallest interval for which the function repeats itself is called:
- a) Periodic function
 - b) Period
 - c) Cosine function
 - d) Sine function
- 3) The range of cosine function is between:
- a) -1 and 2
 - b) 1 and 2
 - c) -1 and 1
 - d) All of these
- 4) $\sin\theta$ is positive in _____ quadrant and negative in _____ quadrant:
- a) 1st, 2nd
 - b) 3rd, 4th
 - c) 2nd, 3rd
 - d) 1st and 2nd, 3rd and 4th
- 5) $\sin\theta =$
- a) $\frac{PQ}{OP}$
 - b) $\frac{OP}{PQ}$
 - c) $\frac{P}{Q}$
 - d) $\frac{O}{P}$
- 6) $\tan\theta =$
- a) $\frac{PQ}{OQ}$
 - b) $\frac{x}{y}$
 - c) $\frac{y}{x}$
 - d) Both a and c
- 7) $\tan\theta$ is positive in _____ quadrant:
- a) 1st and 2nd
 - b) 1st and 3rd
 - c) 1st and 4th
 - d) 2nd and 3rd
- 8) $\tan(180^\circ - \theta) = ?$
- a) $\tan\theta$
 - b) $-\tan\theta$
 - c) Both
 - d) None of these
- 9) The height from the mean value of the function to its maximum or minimum value of a function is its:

- a) Period
 - b) Periodic function
 - c) Range
 - d) Amplitude
- 10) The range of $y = \arcsin x$ is :
- a) $[-\pi$ and $\pi/4]$
 - b) $[-\pi$ and $\pi/2]$
 - c) $[\pi$ and $\pi/4]$
 - d) $[\pi$ and $\pi/2]$
- 11) The domain of $y = \arctan x$ is :
- a) $(-\infty$ and $\infty)$
 - b) $(\infty$ and $0)$
 - c) $(-\pi/2$ and $\pi/2)$
 - d) Both a and b
- 12) The inverse tangent function is defined by $y = \arctan x$ if and only if :
- a) $\tan x = x$
 - b) $\tan y = y$
 - c) $\tan y = x$
 - d) $\tan x = y$
- 13) ☐ The cosine function has period of
- a) 180
 - b) 360
 - c) 240
 - d) 90
- 14) cosine function, has the range between:
- a) -1 and 1.
 - b) 1 and 3
 - c) 1 and 2
 - d) -2 and 1
- 15) For tangent function period is:
- a) 90
 - b) 80
 - c) 180
 - d) 260
- 16) $\sin(\theta - 180) = ?$
- a) $\sin^*$
 - b) $-\sin^*$
 - c) Both
 - d) None
 - e)

17) To solve $\tan \theta = k$

Ans:

___ find $\tan^{-1}(k) = \theta$

___ use the periodic property $\tan(180^\circ + \theta) = \tan(\theta)$ to get the other solution in the required interval.

18) Given that $\tan(63.43^\circ) = 2$, solve $\tan \theta = -2$, giving all the roots in $0^\circ < \theta < 360^\circ$?

Ans :

copy+paste it

19) An equation is not an identity is called a _____ equation:

- a) Identical
- b) Non identical
- c) Conditional
- d) Non conditional

20) In the reciprocal identity $\sec \theta = ?$

- a) $1/\cos \theta$
- b) $1/\sin \theta$
- c) $1/\tan \theta$
- d) $1/\csc \theta$

21) In periodic properties $\tan(\theta + \pi) =$

- a) $\sin \theta$
- b) $\cos \theta$
- c) $\tan \theta$
- d) $\sec \theta$

22) In periodic properties $\sec(\theta + 2\pi) = ?$

- a) $\csc \theta$
- b) $\sec \theta$
- c) $\cot \theta$
- d) $\tan \theta$

23) $\tan^2 \theta + 1 = ?$

a) $\sec^2$

b) $\sin^2$

c) $\cos^2$

d) $\tan^2$

24) $\tan(A-B)=$

a) $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

b) $\frac{\tan A + \tan B}{1 + \tan A \tan B}$

c) $\frac{\tan A + \tan B}{1 - \tan A \tan B}$

d) $\frac{\tan A - \tan B}{1 - \tan A \tan B}$

25) Verify that the following Equation is an identity:

$$\tan t - \cot t / \sin t \cos t = \sec^2 t - \csc^2 t$$

Solution:

$$LHS = RHS$$

$$= \tan t - \cot t / \sin t \cos t = \tan t / \sin t \cos t - \cot t / \sin t \cos t$$

$$= \tan t \cdot \frac{1}{\sin t \cos t} - \cot t \cdot \frac{1}{\sin t \cos t}$$

$$= \sin t / \cos t \cdot \frac{1}{\sin t \cos t} - \cos t / \sin t \cdot \frac{1}{\sin t \cos t}$$

$$= \frac{1}{\cos^2 t} - \frac{1}{\sin^2 t}$$

$$= \sec^2 t - \csc^2 t$$

26) What is double angle formula? prove by sin, cos and tanOs?

$$= 2 \tan A / 1 - \tan^2 A$$

Prove

$$= \sin 2A = \sin(A+A) = \sin A \cos A + \sin A \cos A = 2 \sin A \cos A$$

$$= \cos 2A = \cos(A+A) = \cos A \cos A - \sin A \sin A = \cos^2 A - \sin^2 A$$

$$= \tan 2A = \frac{\sin 2A}{\cos 2A} = \frac{2 \sin A \cos A}{\cos^2 A - \sin^2 A} = \frac{2 \tan A}{1 - \tan^2 A}$$

$$=2\tan A/1-\tan^2 A \text{ proved}$$